1. Data Source and Analytical Framework

This analysis examines temporal, demographic, and geographic trends in reported measles cases in Saudi Arabia between 2014 and 2023, based on surveillance data. Using descriptive statistics and incidence rates, general patterns were identified, and regional variations were assessed through population-adjusted observed-to-expected (O/E) case ratios, multi-group simulations, and a geographic concentration index. Due to data limitations, the analysis of age groups and the monthly distribution at the national level remained restricted.

2. National Monthly Distribution and Temporal Patterns of Reported Measles Cases

The monthly case counts were organized and converted into annual percentages. In the figure below, we display two heatmaps; the first shows the absolute number of cases while the second shows monthly percentages, allowing for a seasonal comparison.
We display these figures side-by-side, making it easy to distinguish the distribution throughout the year (Figure 1).

Monthly Counts

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2014	17	20	17	18	16	10	4	7	5	13	13	14
2015	20	6	4	38	54	51	17	15	5	5	4	0
2016	4	9	8	4	7	6	19	23	6	12	9	18
2017	82	84	39	26	17	8	17	13	28	27	53	63
2018	73	105	185	251	169	73	69	34	39	52	51	55
2019	292	424	157	62	19	26	14	13	1	3	12	1
2020	5	8	12	7	1	0	0	0	0	0	0	1
2021	0	1	2	0	3	3	1	1	67	123	96	30
2022	18	3	6	1	10	33	15	39	40	20	27	35
2023	110	161	400	351	372	243	191	152	96	71	47	60

Monthly Percentages

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2014	11,039	12,987	11,039	11,6883	10,3896	6,4935	2,5974	4,5455	3,2468	8,4416	8,4416	9,0909
2015	9,1324	2,7397	1,8265	17,3516	24,6575	23,2877	7,7626	6,8493	2,2831	2,2831	1,8265	0
2016	3,2	7,2	6,4	3,2	5,6	4,8	15,2	18,4	4,8	9,6	7,2	14,4
2017	17,9431	18,3807	8,5339	5,6893	3,7199	1,7505	3,7199	2,8446	6,1269	5,9081	11,5974	13,7856
2018	6,3149	9,083	16,0035	21,7128	14,6194	6,3149	5,9689	2,9412	3,3737	4,4983	4,4118	4,7578
2019	28,5156	41,4062	15,332	6,0547	1,8555	2,5391	1,3672	1,2695	0,0977	0,293	1,1719	0,0977

2020	14,7059	23,5294	35,2941	20,5882	2,9412	0	0	0	0	0	0	2,9412
2021	0	0,3058	0,6116	0	0,9174	0,9174	0,3058	0,3058	20,4893	37,6147	29,3578	9,1743
2022	7,2874	1,2146	2,4291	0,4049	4,0486	13,3603	6,0729	15,7895	16,1943	8,0972	10,9312	14,17
2023	4,8802	7,1429	17,7462	15,5723	16,504	10,7808	8,4738	6,7436	4,2591	3,15	2,0852	2,6619

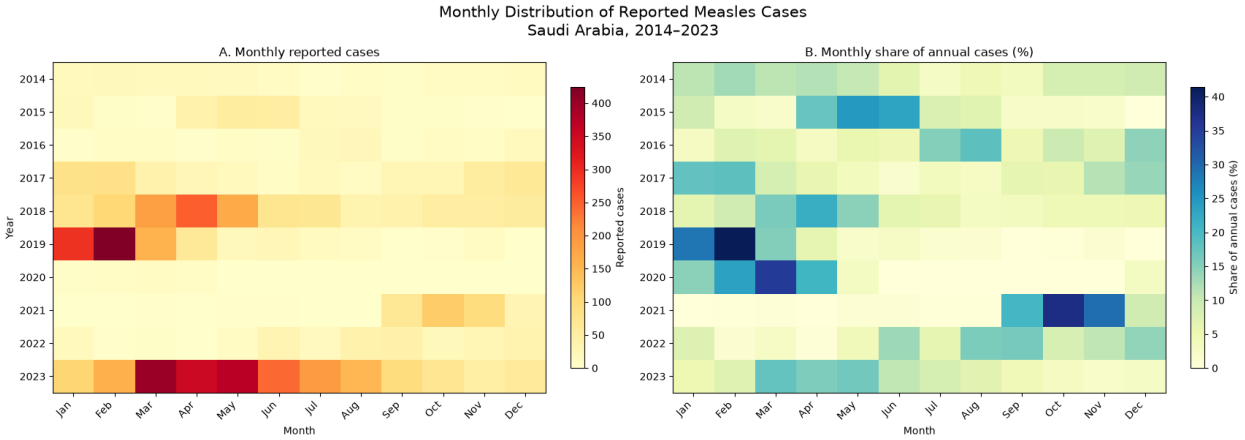


Figure 1

In order to see the intensity of measles cases in a year, we analyse the peak month and the three month peack. Then we calculate the percentages of all cases of that year happened during these period. We then display these figures side by side in a bar chart (Figure 2).

Monthly Distribution of Reported Measles Cases

Year	Annual cases	Peak month	Peak month cases	Peak month share (%)	Peak 3-month period	Peak 3-month cases	Peak 3-month share (%)
2014	154	Feb	20	12,99	Feb–Apr	55	35,71
2015	219	May	54	24,66	Apr–Jun	143	65,3
2016	125	Aug	23	18,4	Jun–Aug	48	38,4
2017	457	Feb	84	18,38	Jan–Mar	205	44,86
2018	1156	Apr	251	21,71	Mar–May	605	52,34
2019	1024	Feb	424	41,41	Jan–Mar	873	85,25
2020	34	Mar	12	35,29	Feb–Apr	27	79,41
2021	327	Oct	123	37,61	Sep–Nov	286	87,46
2022	247	Sep	40	16,19	Aug–Oct	99	40,08
2023	2254	Mar	400	17,75	Mar–May	1123	49,82

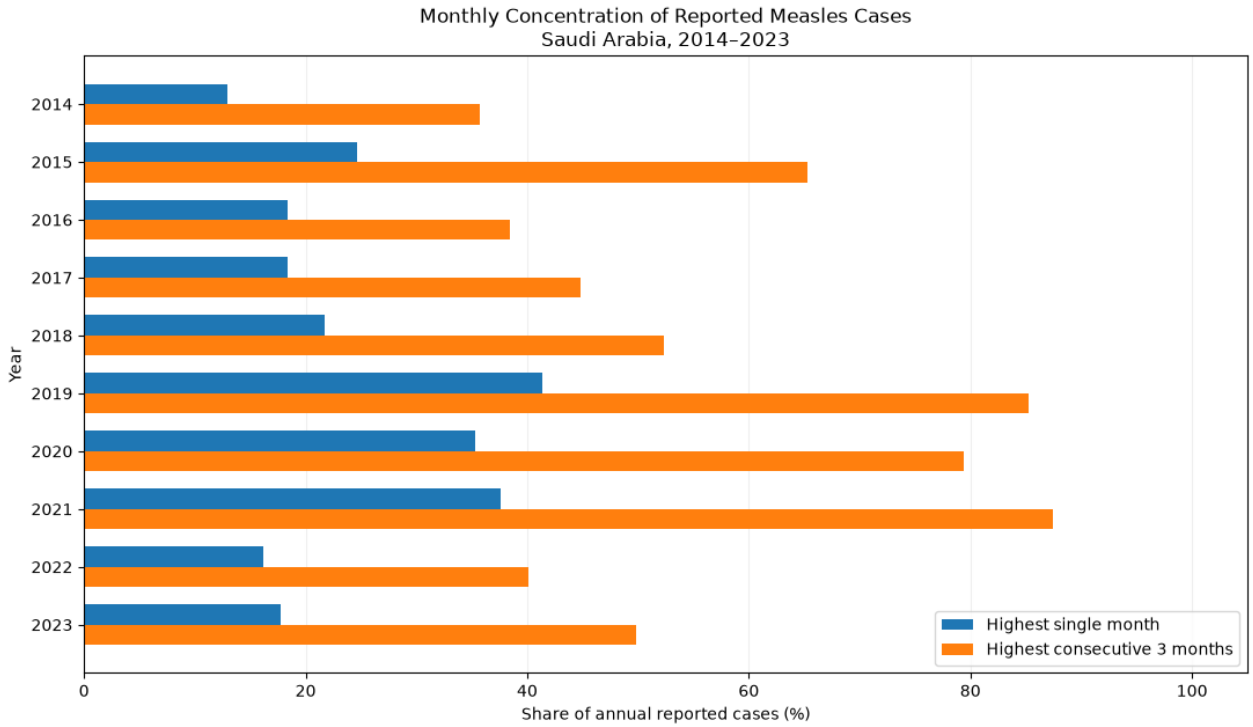

Monthly Concentration of Reported Measles Cases
Saudi Arabia, 2014–2023
Year
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
0
20
40
60
80
100
Share of annual reported cases (%)
Highest single month
Highest consecutive 3 months

Figure 2

3. National Age-Group Distribution

3.1. Annual Age-Group Distribution

Annual measles cases were grouped into five age categories. We calculated the percentage contribution of each group. We then display two stacked bar charts to visualize the case counts and percentage distribution across the study period (Figure 3, 4).

Annual Count

Year	<1 year	1–4 years	5–14 years	15–44 years	45+ years
2014	33	55	26	35	5
2015	65	73	21	55	5
2016	33	43	16	28	5
2017	164	132	58	90	13
2018	356	436	188	170	6
2019	403	391	122	97	11
2020	4	14	12	4	0
2021	94	120	25	86	2
2022	49	88	40	68	2
2023	561	916	565	185	27

Annual Percentage

Year	<1 year	1–4 years	5–14 years	15–44 years	45+ years
2014	21,4286	35,7143	16,8831	22,7273	3,2468
2015	29,6804	33,3333	9,589	25,1142	2,2831
2016	26,4	34,4	12,8	22,4	4
2017	35,8862	28,884	12,6915	19,6937	2,8446
2018	30,7958	37,7163	16,263	14,7059	0,519
2019	39,3555	38,1836	11,9141	9,4727	1,0742
2020	11,7647	41,1765	35,2941	11,7647	0
2021	28,7462	36,6972	7,6453	26,2997	0,6116
2022	19,8381	35,6275	16,1943	27,5304	0,8097
2023	24,8891	40,6389	25,0665	8,2076	1,1979

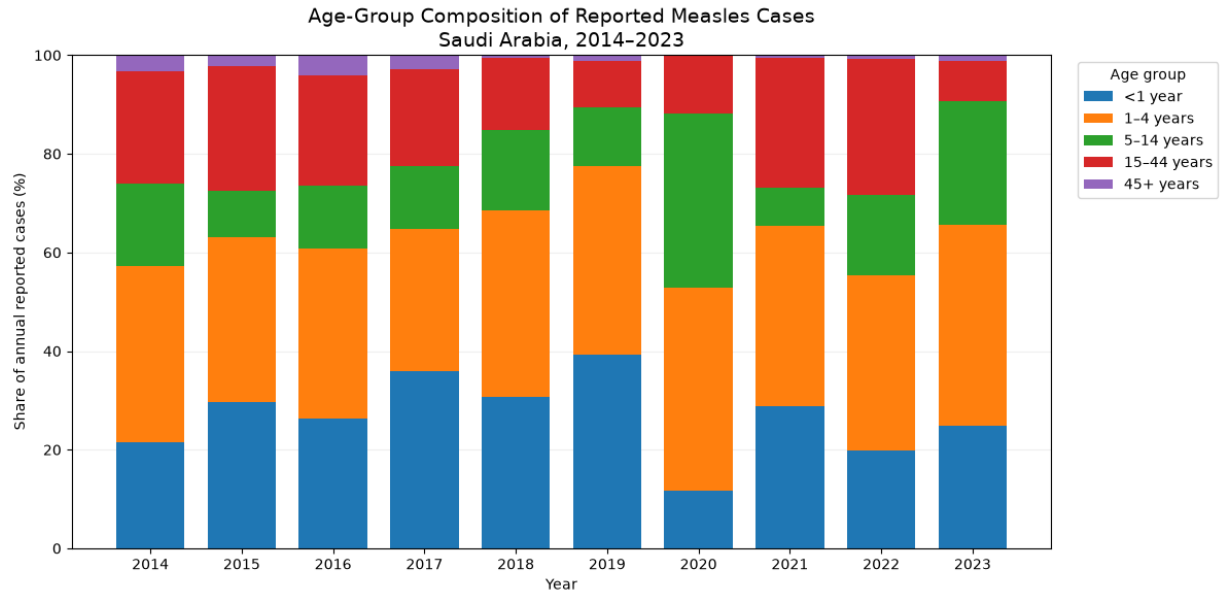


Figure 3

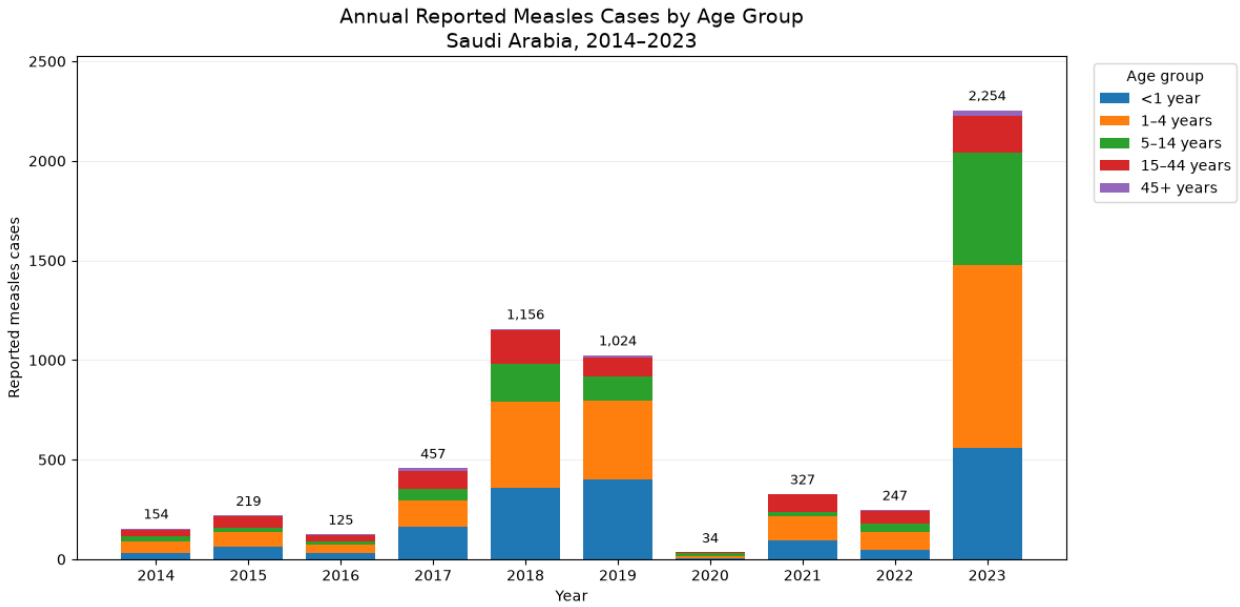


Figure 4

3.2. Age-Group Comparison of Major Measles Reporting Years

We used the three highest case years (2018, 2019, and 2023) to compare age groups and case counts and percentage distributions. We also calculated for each group to quantify its contribution to the overall increase.

Age group	2019 cases	2023 cases	Case difference	Share of total difference (%)
<1 year	403	561	158	12,85
1–4 years	391	916	525	42,68
5–14 years	122	565	443	36,02
15–44 years	97	185	88	7,15
45+ years	11	27	16	1,3

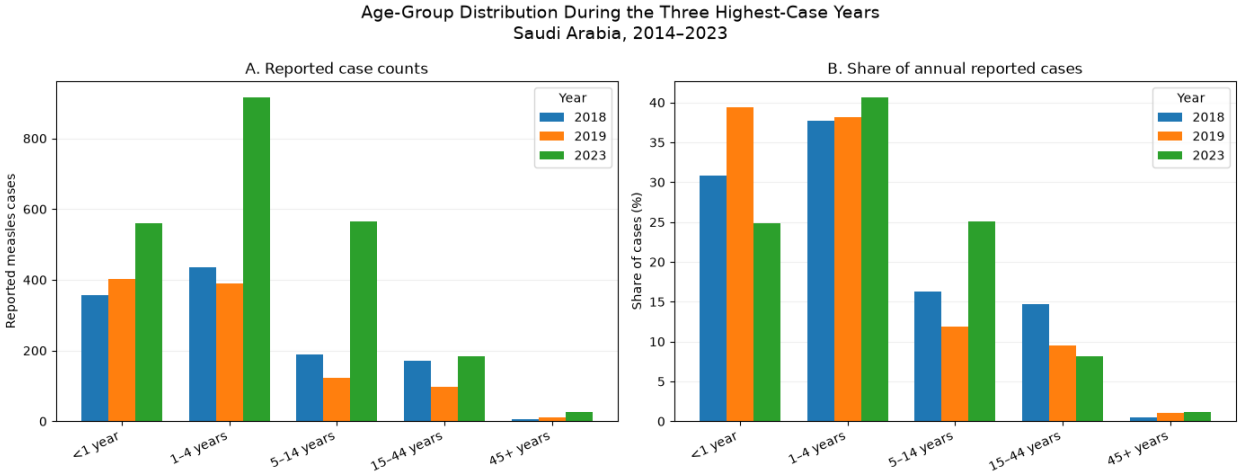


Figure 5

4. Regional and Annual Incidence Heatmap

4.1. Liner heatmap

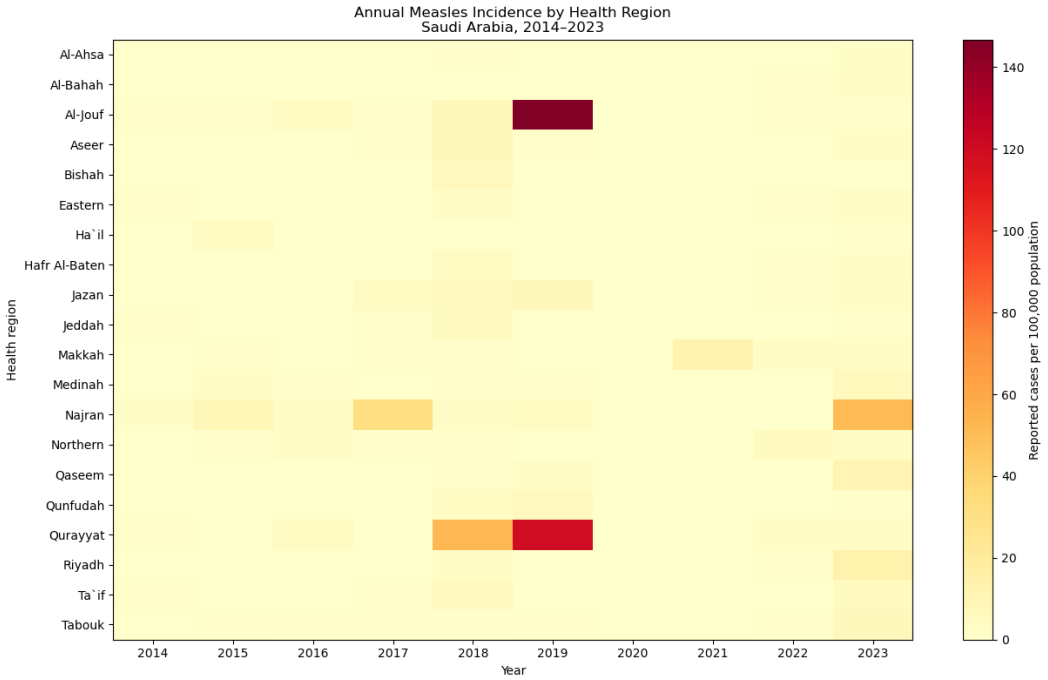


Figure 6

The map is pale and lacks contrast, making it difficult to distinguish meaningful differences among regions with lower incidence rates (Figure 6). Therefore, we use a logarithmic color transformation log1p, defined as: $y = \ln(1 + x)$. (Figure 7)

4.2. Log-transformed heatmap

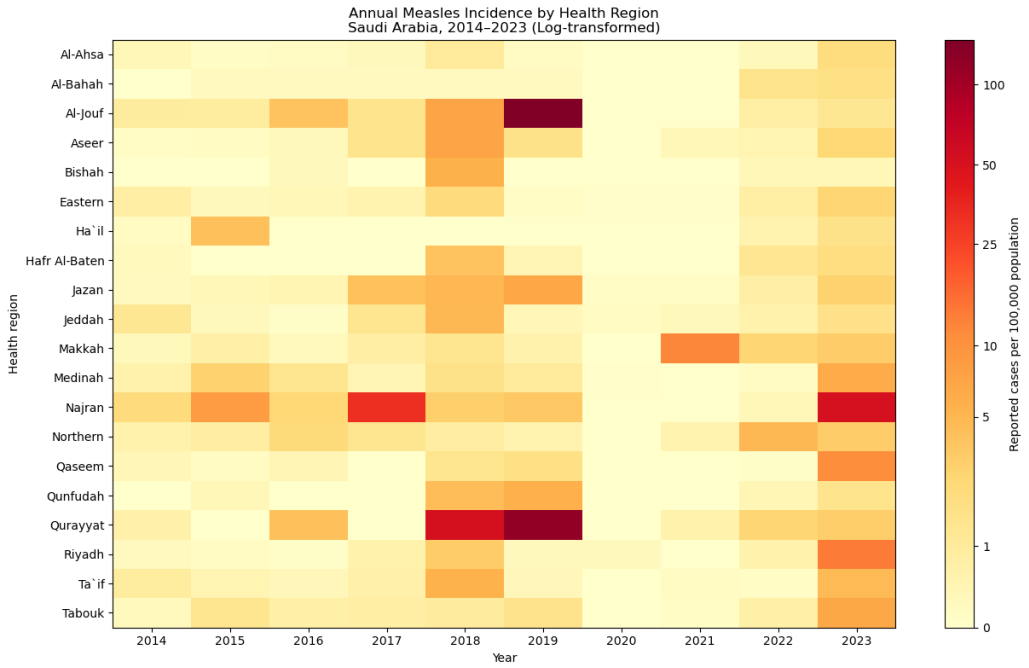


Figure 7

5. Regional Annual Incidence Trajectories

We plotted the incident rates for each one of the 20 health regions (Figure 8).
Annual measles incidence rates were plotted separately for each of the 20 health regions.

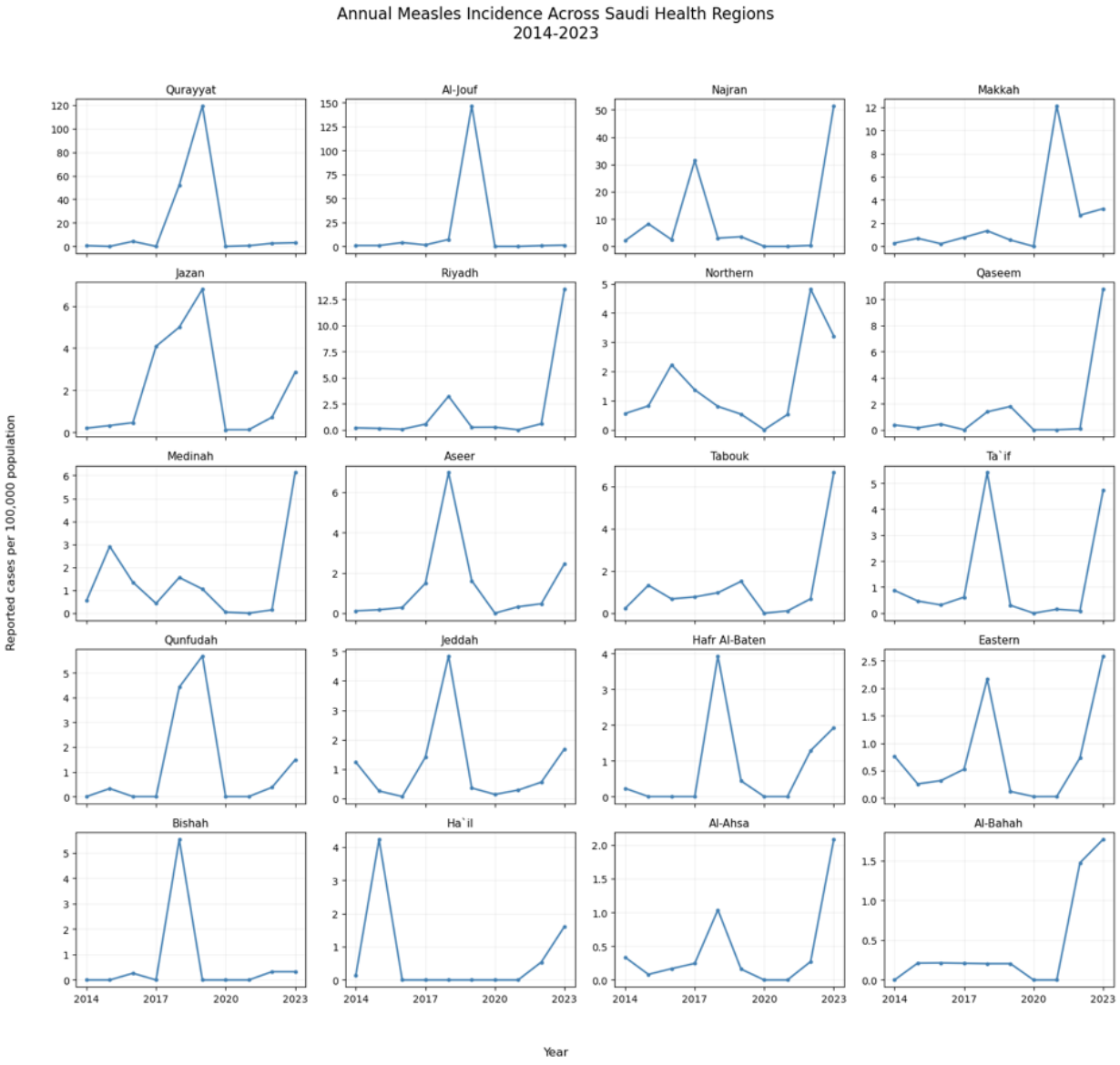


Figure 8

6. Regional Peak-Year Concentration

We identified the year with the highest reported cases for each health region. Then we devided the peak year case count by the region's total cases from 2014 to 2023. We then calculated the concentration percentage. Thn we displayed it in a bar chart (figure 9).

Health region	Peak year	Peak year cases	Total cases	Peak year share percentage
Al-Jouf	2019	510	567	89,95%
Bishah	2018	22	25	88,00%
Riyadh	2023	1157	1600	72,31%
Qaseem	2023	144	204	70,59%
Qurayyat	2019	207	317	65,30%
Ha`il	2015	29	46	63,04%
Makkah	2021	297	545	54,50%
Aseer	2018	130	253	51,38%
Tabouk	2023	59	116	50,86%
Najran	2023	305	603	50,58%
Hafr Al-Baten	2018	18	36	50,00%
Qunfudah	2019	18	38	47,37%
Jeddah	2018	231	489	47,24%
Al-Ahsa	2023	23	51	45,10%
Ta`if	2018	72	160	45,00%
Medinah	2023	131	300	43,67%
Al-Bahah	2023	6	16	37,50%
Eastern	2023	92	254	36,22%
Jazan	2019	109	322	33,85%
Northern	2022	18	55	32,73%

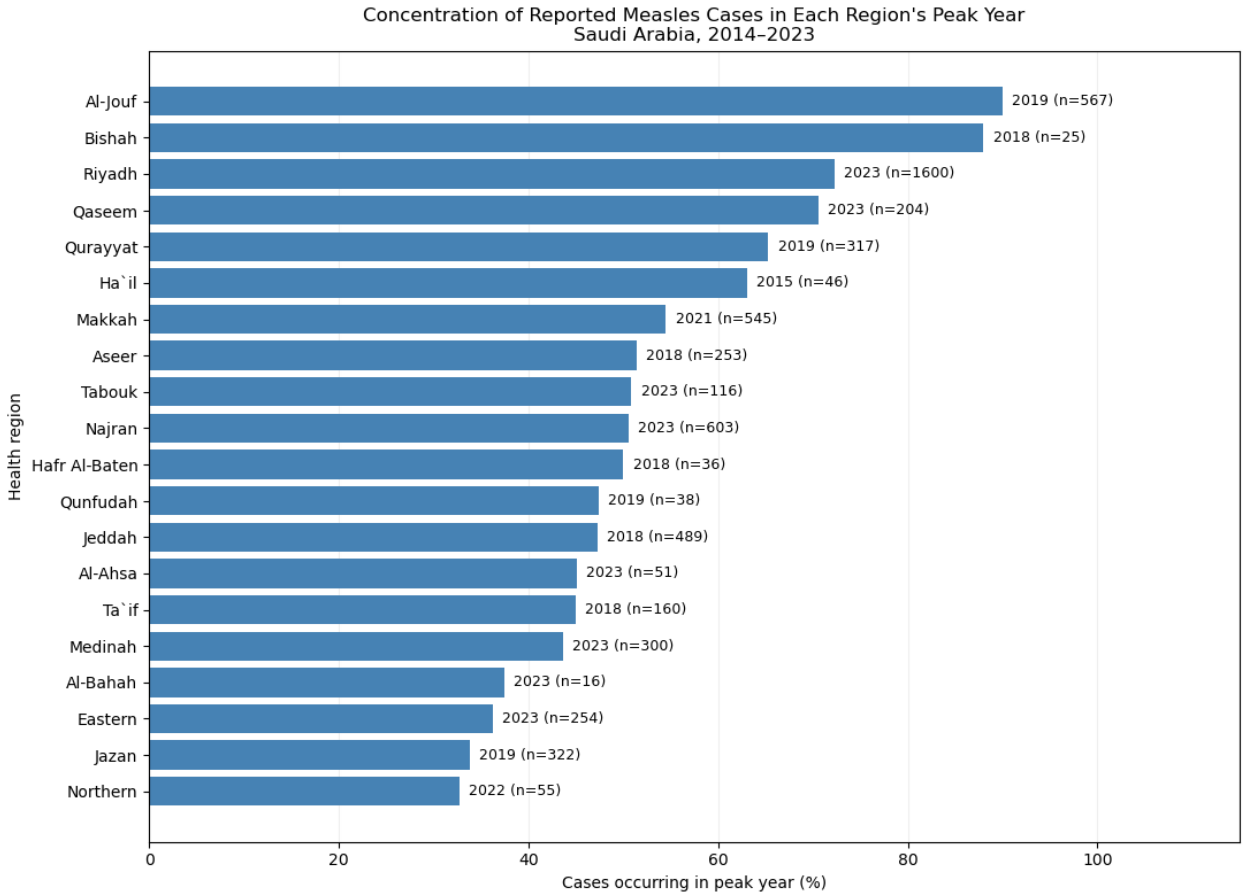

Concentration of Reported Measles Cases in Each Region's Peak Year
Saudi Arabia, 2014–2023
Health region
Al-Jouf
Bishah
Riyadh
Qaseem
Qurayyat
Ha`il
Makkah
Aseer
Tabouk
Najran
Hafr Al-Baten
Qunfudah
Jeddah
Al-Ahsa
Ta`if
Medinah
Al-Bahah
Eastern
Jazan
Northern
2019 (n=567)
2018 (n=25)
2023 (n=1600)
2023 (n=204)
2019 (n=317)
2015 (n=46)
2021 (n=545)
2018 (n=253)
2023 (n=116)
2023 (n=603)
2018 (n=36)
2019 (n=38)
2018 (n=489)
2023 (n=51)
2018 (n=160)
2023 (n=300)
2023 (n=16)
2023 (n=254)
2019 (n=322)
2022 (n=55)
0
20
40
60
80
100
Cases occurring in peak year (%)

Figure 9

7. Geographic Composition of Major Measles Reporting Years

We selected the highest three years with case counts (2018, 2019, 2023). We then calculate the percentage contribution of each region to the national total. We then viulised it in a stack bar(Figure 10).

Regional Contributions to National Measles Cases in 2018, 2019, and 2023 (%)

Year	Riyadh	Jeddah	Aseer	Qurayyat	Jazan	Al-Jouf	Qaseem	Najran	Medinah	Eastern	Other regions
2018	23,5294	19,9827	11,2457	7,7855	6,9204	2,1626	1,7301	1,5571	2,9412	6,2284	15,917
2019	2,0508	1,6602	2,9297	20,2148	10,6445	49,8047	2,5391	2,0508	2,2461	0,3906	5,4688
2023	51,331	2,9725	1,8634	0,2662	1,7746	0,2218	6,3886	13,5315	5,8119	4,0816	11,7569

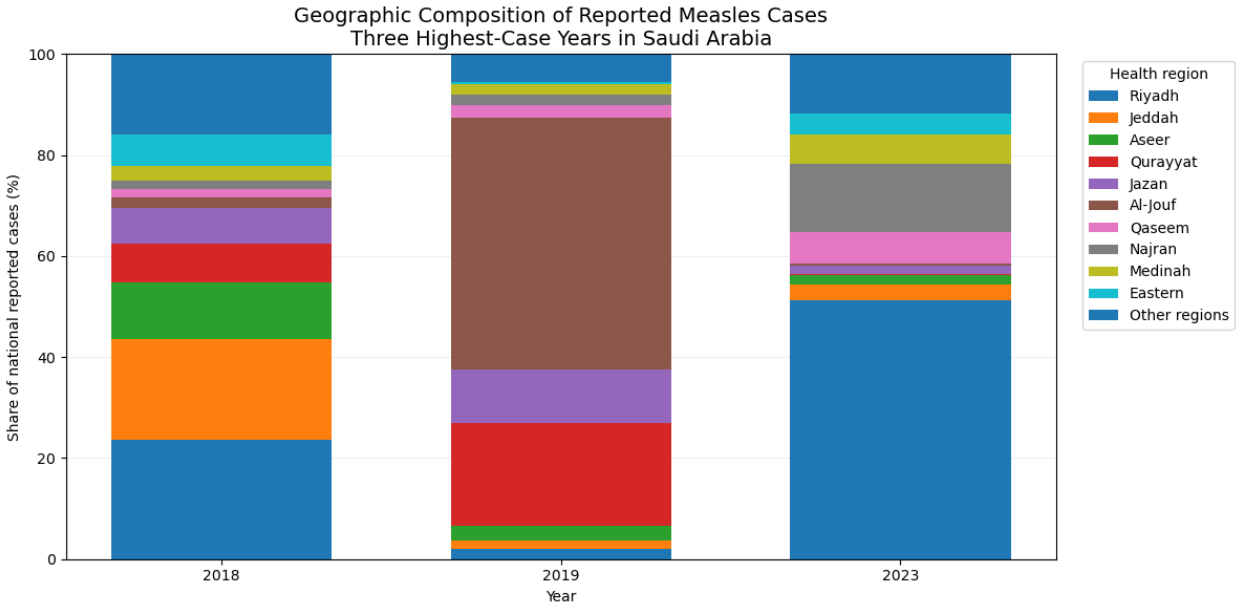


Figure 10

8. Observed-to-expected (O/E) ratio

$$O/E_r = \frac{O_r}{E_r}$$

Where:

O_r: Total observed cases in region r over 2014-2023

E_r: Total expected cases, calculated using the corresponding national annual incidence rates and that region's annual populations.

$$E_r = \sum_{t=2014}^{2023} (\frac{C_t}{P_t} \times P_{r,t})$$

C_t is the national case count, P_t is the national population, and $P_{r,t}$ is the regional population in year t.

For each health region, the total observed measles cases during 2014–2023 were compared with the expected cases, calculated using annual national incidence rates and corresponding regional populations. The observed-to-expected (O/E) ratio was then calculated to assess regional differences relative to the national reference. The 95% confidence interval (CI) provides a range of plausible values for the O/E ratio. The lower and upper limits define this range and reflect uncertainty in the estimated ratio (Figure 11).

Observed-to-Expected Measles Cases and 95% Confidence Intervals by Health Region (2014–2023)

Health region	Observed cases	Expected cases	Total population exposure	O/E ratio	95% CI lower (source)	95% CI upper (source)
Qurayyat	317	33.365	1,763,900	9.501	8.484	10.607
Al-Jouf	567	67.631	3,551,052	8.384	7.708	9.103
Najran	603	108.627	5,906,522	5.551	5.117	6.012
Makkah	545	456.722	23,919,901	1.193	1.095	1.298
Jazan	322	277.746	15,528,022	1.159	1.036	1.293
Riyadh	1600	1,550.511	83,417,447	1.032	0.982	1.084
Northern	55	68.465	3,722,828	0.803	0.605	1.046
Qaseem	204	256.654	14,181,605	0.795	0.690	0.912
Aseer	253	328.866	18,168,190	0.769	0.677	0.870
Medinah	300	395.170	21,473,746	0.759	0.676	0.850
Ta`if	160	225.420	12,735,124	0.710	0.604	0.829
Qunfudah	38	54.343	3,056,697	0.699	0.495	0.960
Tabouk	116	166.649	9,147,824	0.696	0.575	0.835
Jeddah	489	805.627	45,118,915	0.607	0.554	0.663
Hafr Al-Baten	36	84.397	4,563,071	0.427	0.299	0.591
Eastern	254	622.873	33,105,189	0.408	0.359	0.461

Bishah	25	65.910	3,775,568	0.379	0.245	0.560
Ha`il	46	133.105	7,143,656	0.346	0.253	0.461
Al-Ahsa	51	217.426	12,132,673	0.235	0.175	0.308
Al-Bahah	16	77.494	4,531,270	0.206	0.118	0.335

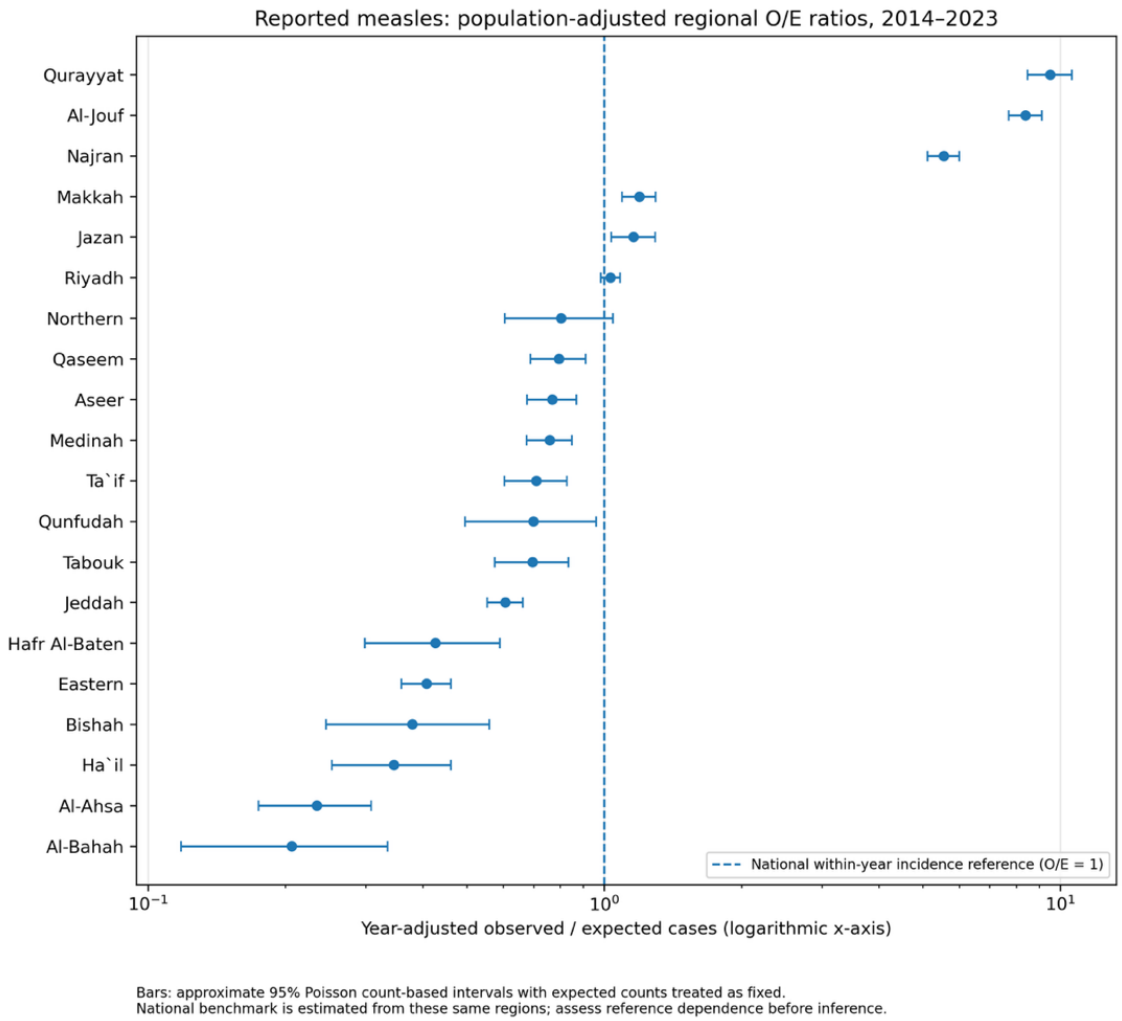


Figure 11

8.1. Sensitivity Analysis of Regional O/E Ratios

Regional O/E ratios were recalculated after separately excluding the peak years (2018, 2019, and 2023). Both observed and expected cases were recomputed for each scenario to assess how individual high-case years influenced the regional comparisons (Figure 12).

Regional O/E Ratios Following Exclusion of Major Reporting

Health region	Total cases	Population exposure	Incidence per 100k	Population share	Case share	Total cases
Al-Bahah	16	4531270	0,353102	0,013860	0,002668	Al-Bahah
Al-Ahsa	51	12132673	0,420353	0,037109	0,008504	Al-Ahsa

Ha`il	46	7143656	0,643928	0,021850	0,007671	Ha`il
Bishah	25	3775568	0,662152	0,011548	0,004169	Bishah
Eastern	254	33105189	0,767251	0,101257	0,042355	Eastern
Hafr Al-Baten	36	4563071	0,788942	0,013957	0,006003	Hafr Al-Baten
Jeddah	489	45118915	1,083803	0,138002	0,081541	Jeddah
Qunfudah	38	3056697	1,243172	0,009349	0,006337	Qunfudah
Ta`if	160	12735124	1,256368	0,038952	0,026680	Ta`if
Tabouk	116	9147824	1,268061	0,027980	0,019343	Tabouk
Aseer	253	18168190	1,392544	0,055570	0,042188	Aseer
Medinah	300	21473746	1,397055	0,065680	0,050025	Medinah
Qaseem	204	14181605	1,438483	0,043376	0,034017	Qaseem
Northern	55	3722828	1,477372	0,011387	0,009171	Northern
Riyadh	1600	83417447	1,918064	0,255144	0,266800	Riyadh
Jazan	322	15528022	2,073670	0,047495	0,053694	Jazan
Makkah	545	23919901	2,278438	0,073162	0,090879	Makkah
Najran	603	5906522	10,209054	0,018066	0,100550	Najran
Al-Jouf	567	3551052	15,967099	0,010861	0,094547	Al-Jouf
Qurayyat	317	1763900	17,971540	0,005395	0,052860	Qurayyat



Figure 12

9. Simulation-based O/E reference visualization

We generated 20,000 multinomial simulations. We kept the annual national case totals fixed and distributing cases across regions according to their population shares. The resulting O/E ratios were used to calculate 95% null reference intervals, representing the variation expected under equal population adjusted incidence across regions (Figure 13).

Regional O/E Ratios and 95% Simulation-Based Null Reference Intervals (2014–2023)

Health region	95% null lower	95% null upper	Observed O/E ratio
Al-Ahsa	0.869260	1.131418	0.234562
Al-Bahah	0.787161	1.225906	0.206468
Al-Jouf	0.768884	1.242043	8.383791
Aseer	0.897021	1.103792	0.769310
Bishah	0.773783	1.244121	0.379305
Eastern	0.926352	1.074055	0.407788
Ha`il	0.833925	1.172003	0.345591
Hafr Al-Baten	0.793866	1.220421	0.426555
Jazan	0.885703	1.116129	1.159334
Jeddah	0.935917	1.066250	0.606981
Makkah	0.913028	1.088189	1.193286
Medinah	0.905940	1.095732	0.759168
Najran	0.819321	1.187555	5.551127
Northern	0.774121	1.241515	0.803333
Qaseem	0.880563	1.118335	0.794845
Qunfudah	0.736065	1.269713	0.699262
Qurayyat	0.689346	1.348721	9.500987
Riyadh	0.957104	1.043527	1.031918
Ta`if	0.873923	1.131220	0.709785
Tabouk	0.852090	1.152122	0.696074

Note: The 95% null reference bounds were generated from 20,000 population-proportional multinomial simulations. These are not confidence intervals for the observed O/E ratios. Values are rounded to three decimals.

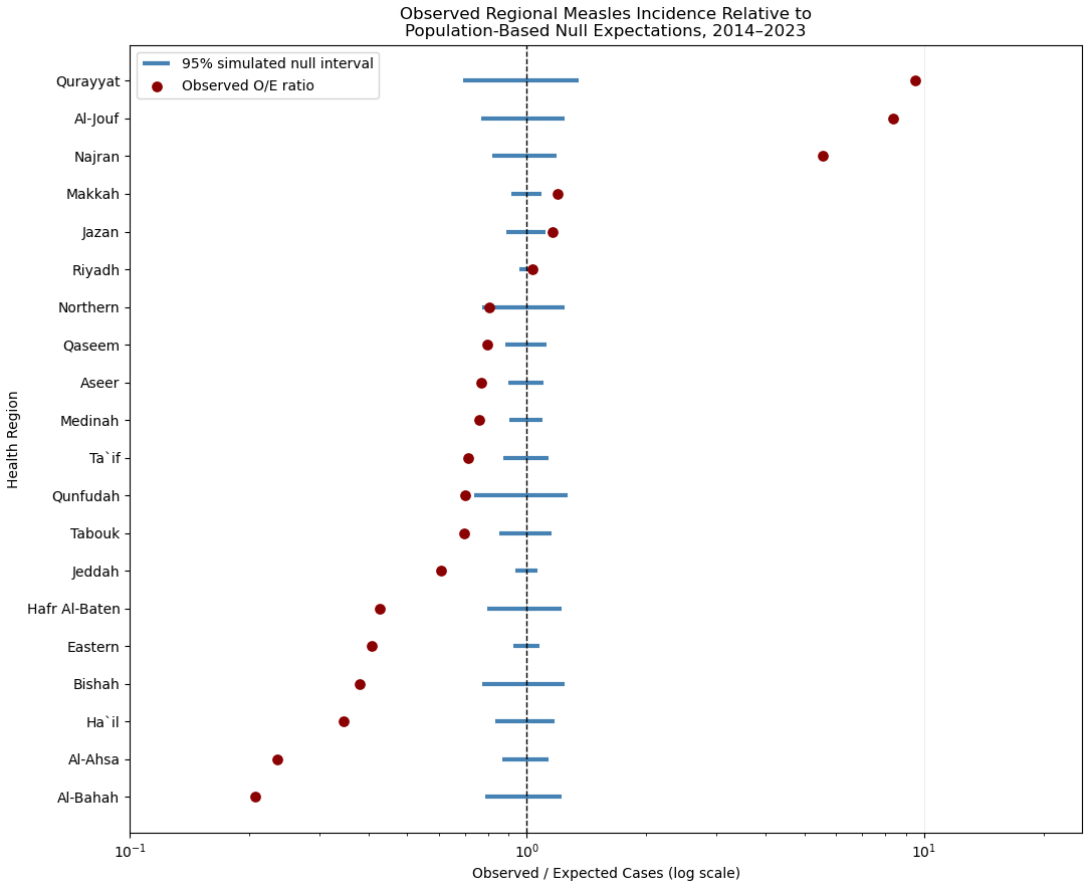

Observed Regional Measles Incidence Relative to
Population-Based Null Expectations, 2014–2023
95% simulated null interval
Observed O/E ratio
Qurayyat
Al-Jouf
Najran
Makkah
Jazan
Riyadh
Northern
Qaseem
Aseer
Medinah
Ta`if
Qunfudah
Tabouk
Jeddah
Hafr Al-Baten
Eastern
Bishah
Ha`il
Al-Ahsa
Al-Bahah
Health Region
10^{-1}
10^{0}
10^{1}
Observed / Expected Cases (log scale)

Figure 13

10. Simultaneous Simulation-based reference intervals

To account for comparisons across all 20 health regions, we calculated simultaneous 95% null reference intervals using the maximum standardized deviation in each simulation (Figure 14).

$$T_s = \max_r \left| \frac{O_r^{(s)} - E_r}{\sigma_r} \right|$$

Were:

$O_r^{(s)}$: Simulated cases in region r, simulation s.

E_r: Expected cases in that region

σ_r: Standard deviation under our multinomial reference model

T_s: Largest stander deviation across all 20 regions in one simulation

Health region	Observed cases	Expected cases	O/E ratio	Null SD	Simultaneous lower 95	Simultaneous upper 95
Qurayyat	317	33,36495	9,500987	5,760056	0,476706	1,523294
Al-Jouf	567	67,63051	8,383791	8,176907	0,633515	1,366485
Najran	603	108,6266	5,551127	10,32756	0,711815	1,288185
Makkah	545	456,7219	1,193286	20,53587	0,863708	1,136292
Jazan	322	277,7456	1,159334	16,27424	0,822392	1,177608
Riyadh	1600	1550,511	1,031918	33,90162	0,933724	1,066276
Northern	55	68,46475	0,803333	8,226961	0,635765	1,364235
Qaseem	204	256,6539	0,794845	15,67363	0,814889	1,185111
Aseer	253	328,8662	0,76931	17,6301	0,837503	1,162497
Medinah	300	395,1696	0,759168	19,21272	0,852628	1,147372
Ta`if	160	225,4204	0,709785	14,72748	0,801964	1,198036
Qunfudah	38	54,343	0,699262	7,338162	0,590689	1,409311
Tabouk	116	166,649	0,696074	12,72861	0,76848	1,23152
Jeddah	489	805,6269	0,606981	26,39851	0,900676	1,099324
Hafr Al-Baten	36	84,39708	0,426555	9,121865	0,672383	1,327617
Eastern	254	622,8733	0,407788	23,62186	0,885046	1,114954
Bishah	25	65,90998	0,379305	8,073257	0,628715	1,371285
Ha`il	46	133,1055	0,345591	11,40819	0,740205	1,259795
Al-Ahsa	51	217,4263	0,234562	14,47504	0,798202	1,201798
Al-Bahah	16	77,49369	0,206468	8,744595	0,657955	1,342045

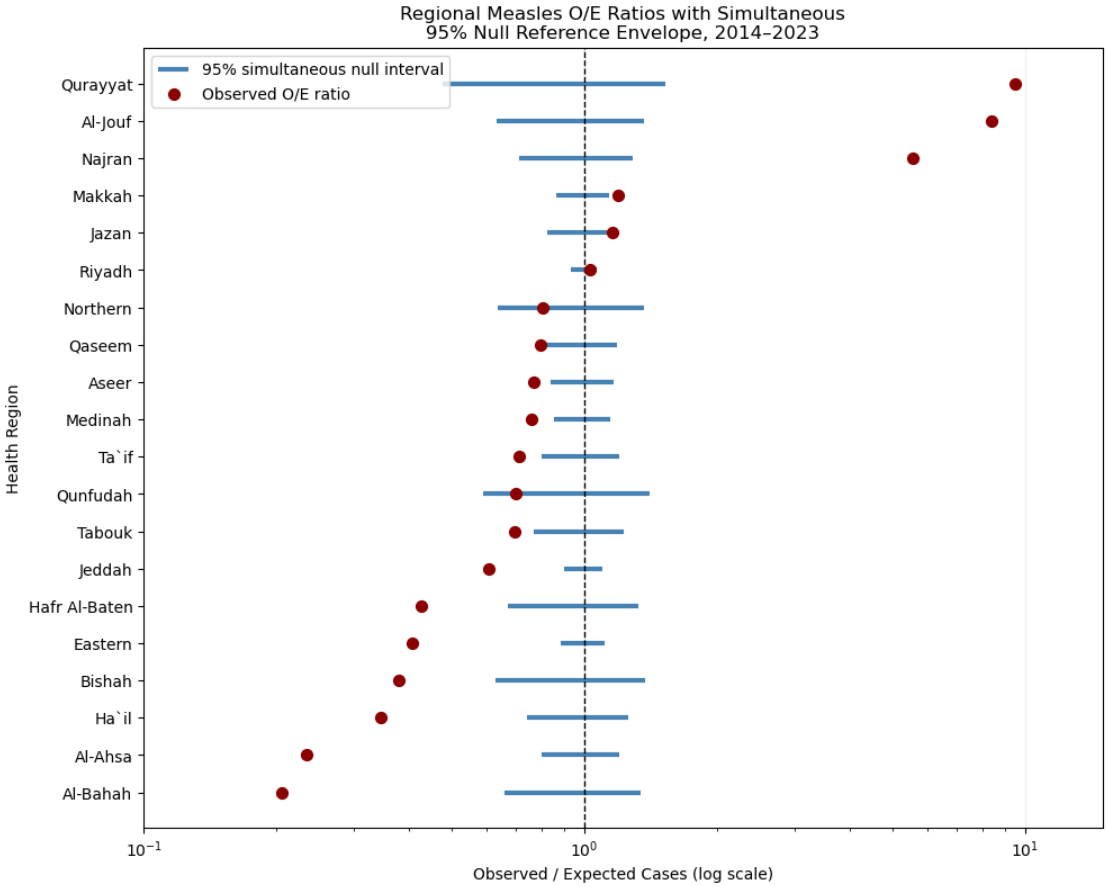


Figure 14

```
Empirical simultaneous coverage: 0.950
Minimum lower O/E limit: 0.47670599335743474
Independent simultaneous coverage: 0.950
Minimum SD ratio: 0.989
Maximum SD ratio: 1.01
SDs agree within 5%: True
Simulations at least as concentrated: 0
Monte Carlo upper-tail p-value: 0.000050
```

We ranked the regions by average annual incidence, and cumulative shares of population exposure and reported cases were plotted. The dashed diagonal indicates proportional equality, while the shaded area represents geographic concentration. The observed Gini-type concentration index was 0.369. (Figure 15).

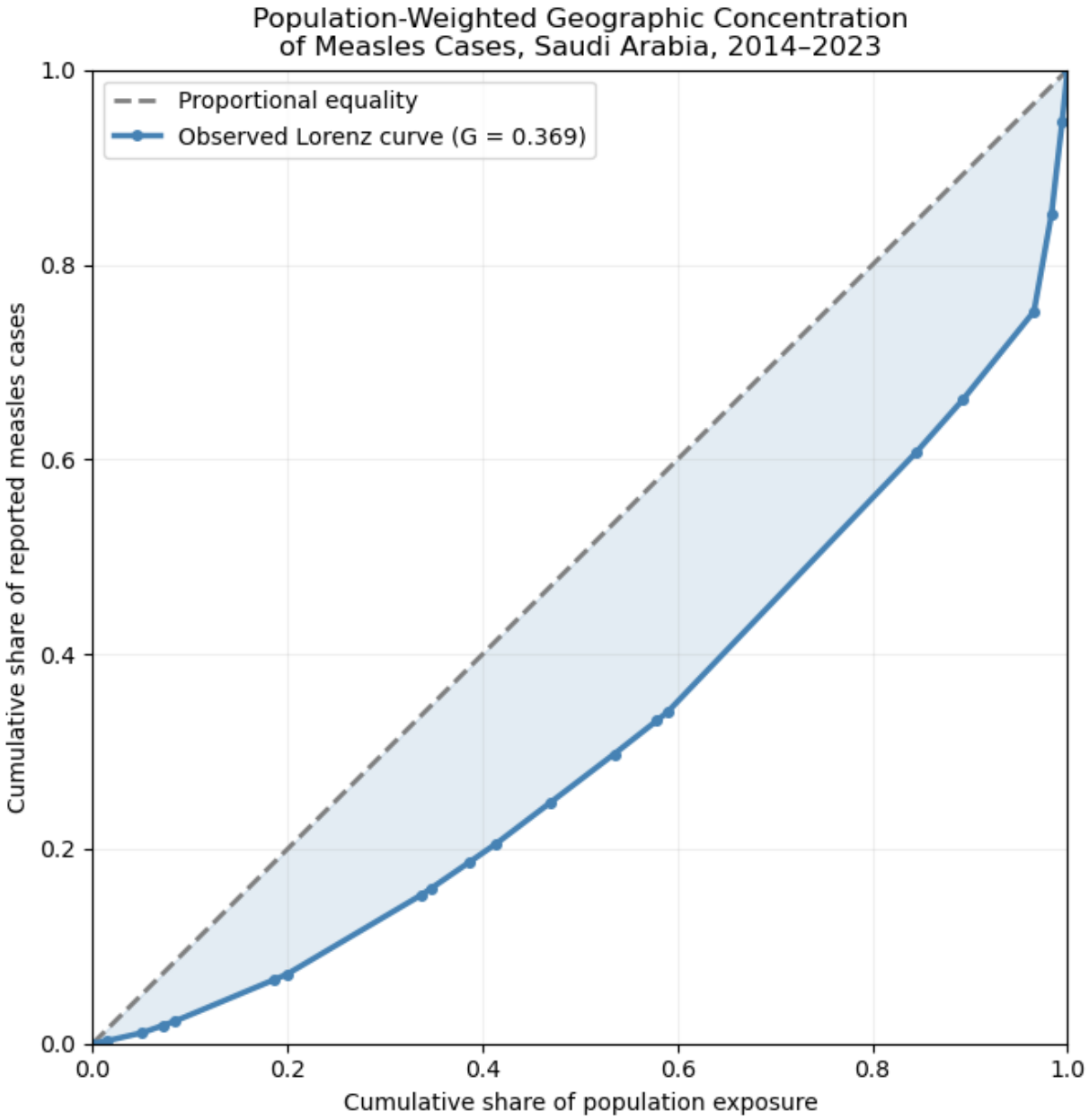


Figure 15 The population-weighted Lorenz curve of reported measles cases across 20 Saudi health regions, 2014–2023

11. Population-weighted geographic concentration

In order to analise the geographic concentration, we ranked health regions by their annoual incidence rates. We used cumulative population and case shares to construct a Lorenz curve and calculate a Gini-type concentration index. The observed index was compared with results from 20,000 population-proportional simulations (Figure 16).

Regional Measles Cases, Population Shares, and Incidence Rates

Health region	Total cases	Population exposure	Incidence per 100k	Population share	Case share
Al-Bahah	16	4531270	0,353102	0,013860	0,002668
Al-Ahsa	51	12132673	0,420353	0,037109	0,008504
Ha`il	46	7143656	0,643928	0,021850	0,007671
Bishah	25	3775568	0,662152	0,011548	0,004169
Eastern	254	33105189	0,767251	0,101257	0,042355
Hafr Al-Baten	36	4563071	0,788942	0,013957	0,006003
Jeddah	489	45118915	1,083803	0,138002	0,081541
Qunfudah	38	3056697	1,243172	0,009349	0,006337
Ta`if	160	12735124	1,256368	0,038952	0,026680
Tabouk	116	9147824	1,268061	0,027980	0,019343
Aseer	253	18168190	1,392544	0,055570	0,042188
Medinah	300	21473746	1,397055	0,065680	0,050025
Qaseem	204	14181605	1,438483	0,043376	0,034017
Northern	55	3722828	1,477372	0,011387	0,009171
Riyadh	1600	83417447	1,918064	0,255144	0,266800
Jazan	322	15528022	2,073670	0,047495	0,053694
Makkah	545	23919901	2,278438	0,073162	0,090879
Najran	603	5906522	10,209054	0,018066	0,100550
Al-Jouf	567	3551052	15,967099	0,010861	0,094547
Qurayyat	317	1763900	17,971540	0,005395	0,052860

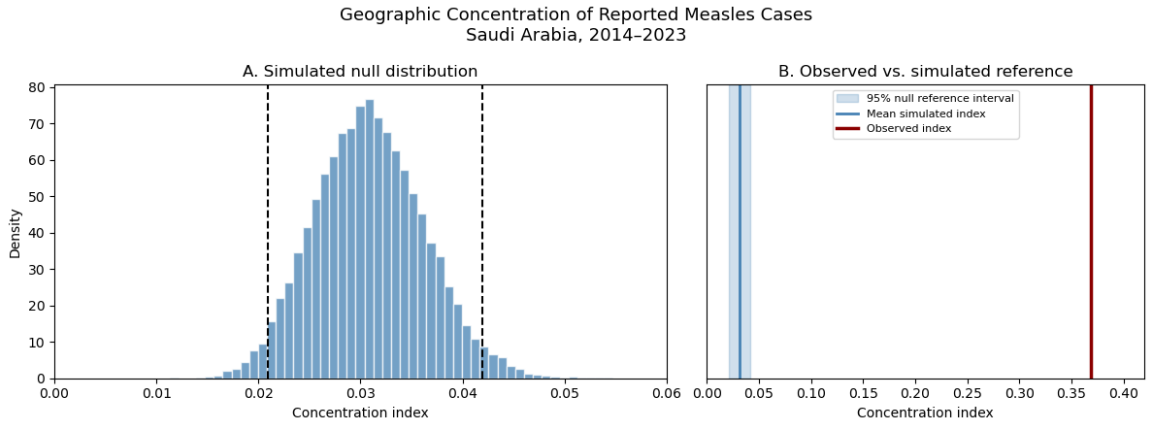


Figure 16

12. Geospatial Visualization

Regional measles incidence was visualized using choropleth maps of the 20 health regions. The visualizations is include a ten year average annual incidence map (Figure 17), annual maps with a consistent color scale (Figure 18), and an interactive map for exploring yearly changes (Figure 19). Provisional health-region boundaries were reconstructed from governorate-level geographic data.

12.1. Geospatial Visualization of Regional Measles Incidence

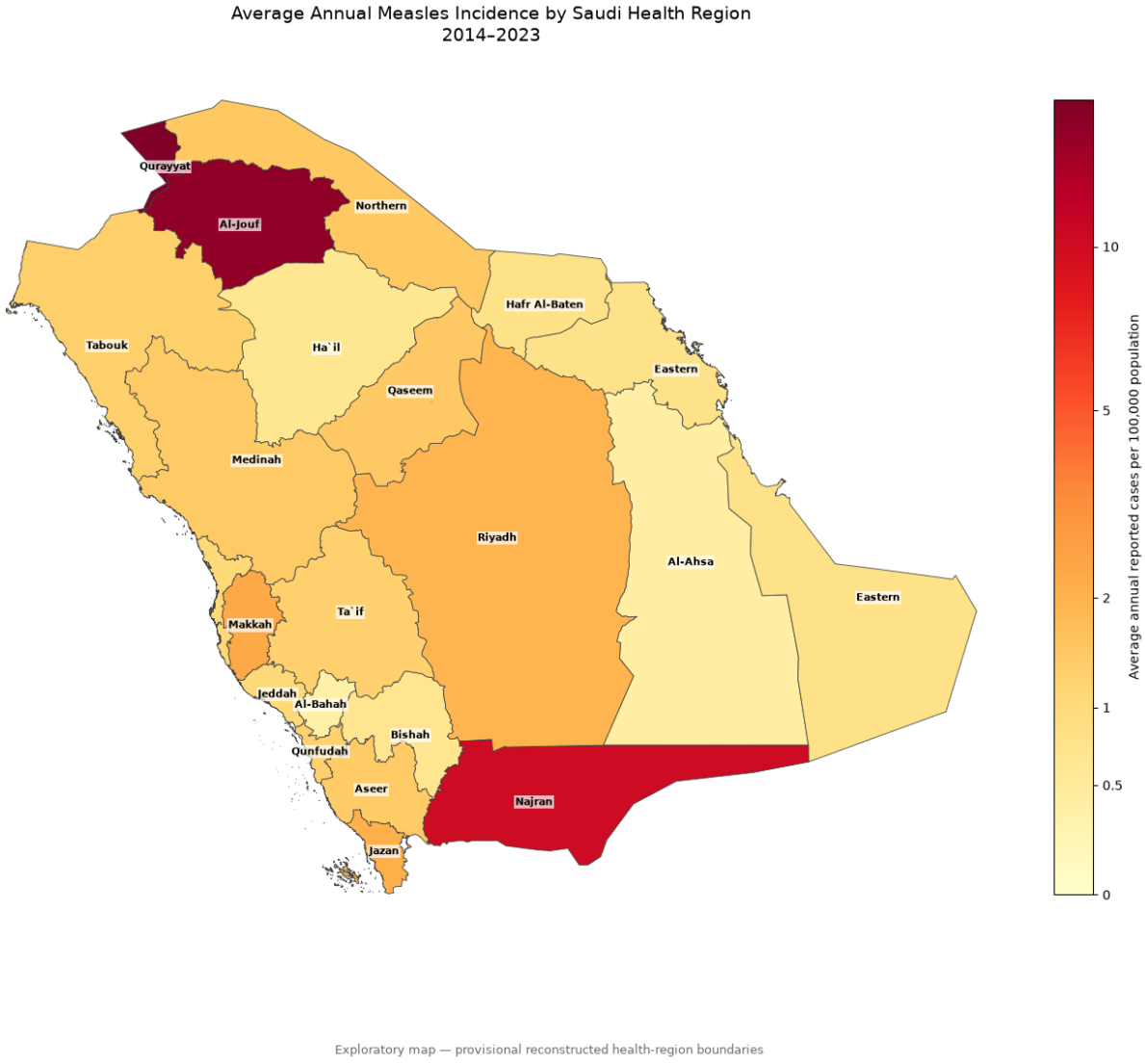


Figure 17

12.2. Geospatial Visualization of Regional Measles Incidence

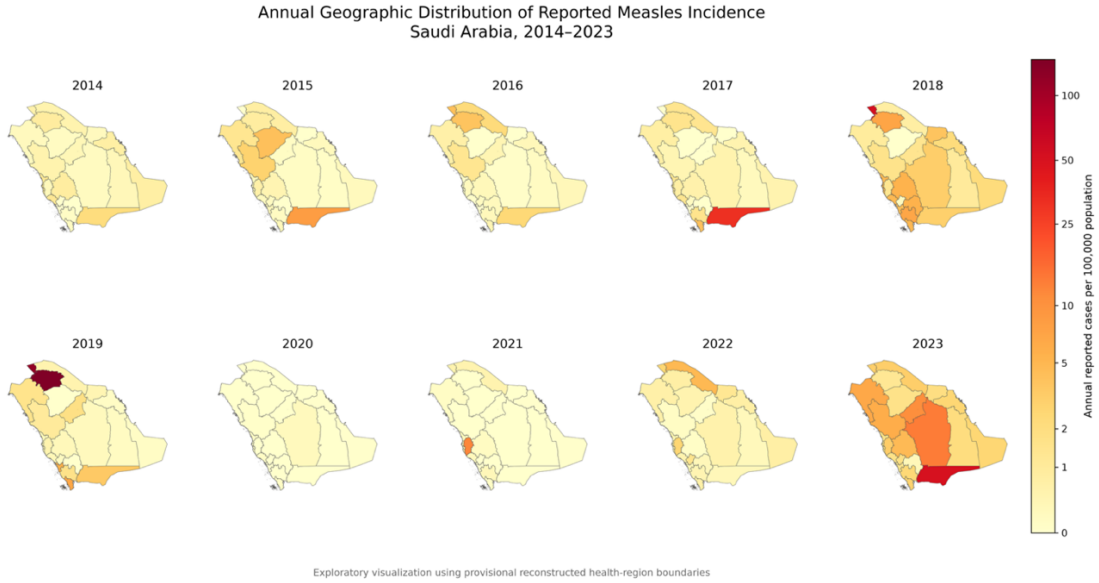


Figure 18

12.3. Interactive annual geographic incident map

Click here to view the interactive map

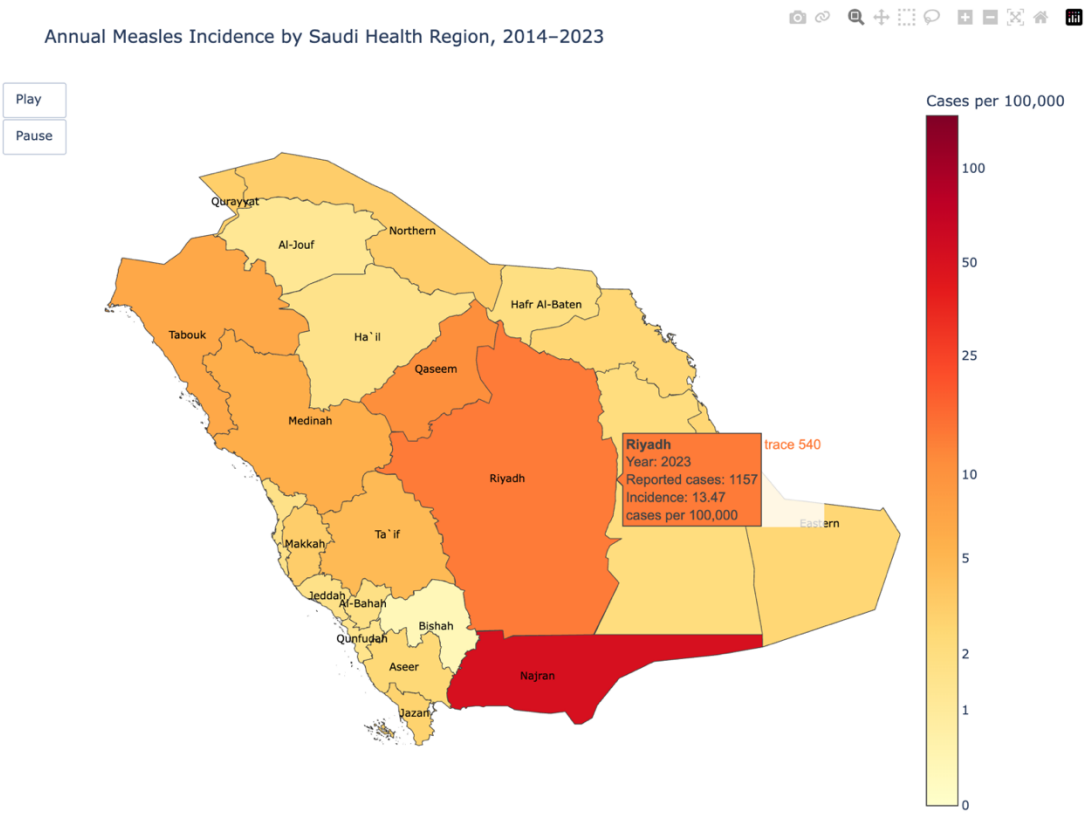


Figure 19

13. Geographic Distribution of Observed-to-Expected Ratios

Regional observed-to-expected (O/E) ratios were displayed on a choropleth map using a diverging color scale. Values above 1 indicate more reported cases than expected, while values below 1 indicate fewer. A logarithmic transformation was applied to improve the visualization of differences between regions (Figure 20).

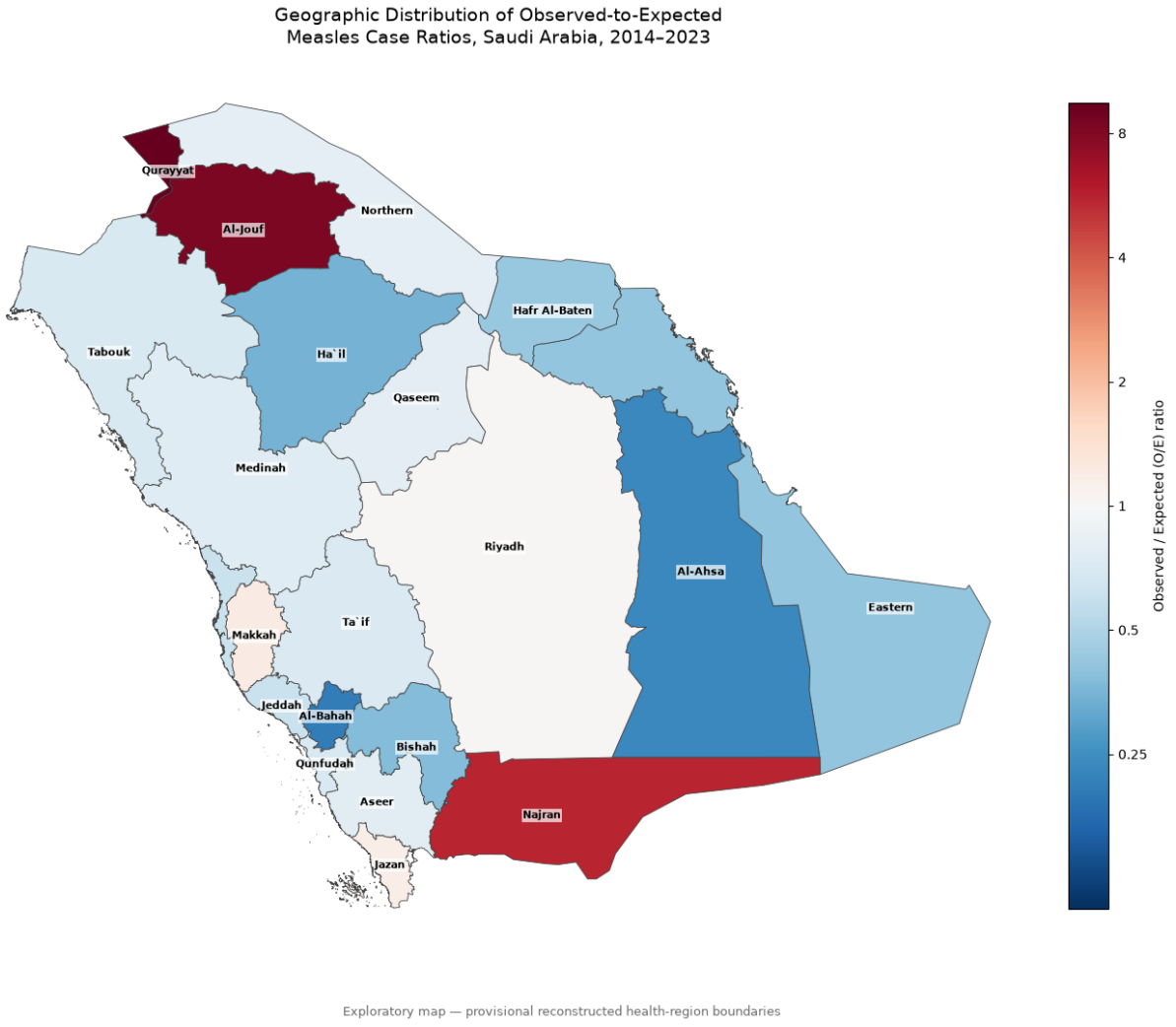


Figure 20

14. Geographic Classification of O/E Deviations

We classifed the health regions as abouve, within, or below the simulataneous 95% null reference base on their O/E ratios. We then displayed the classifcations on a map to highlight the regional deviations from the population-adjusted null expectation (Figure 21).

Classification of Health Regions Based on Simultaneous 95% O/E Null Reference Intervals

Health region	Observed Cases	Expected Cases	O/E Ratio	Simultaneous 95% Null Lower	Simultaneous 95% Null Upper	Classification
Qurayyat	317	33,36495	9,500987	0,476706	1,523294	Higher than null reference
Al-Jouf	567	67,63051	8,383791	0,633515	1,366485	Higher than null reference
Najran	603	108,6266	5,551127	0,711815	1,288185	Higher than null reference
Makkah	545	456,7219	1,193286	0,863708	1,136292	Higher than null reference
Jazan	322	277,7456	1,159334	0,822392	1,177608	Within null reference
Riyadh	1600	1550,511	1,031918	0,933724	1,066276	Within null reference
Northern	55	68,46475	0,803333	0,635765	1,364235	Within null reference
Qaseem	204	256,6539	0,794845	0,814889	1,185111	Lower than null reference
Aseer	253	328,8662	0,76931	0,837503	1,162497	Lower than null reference
Medinah	300	395,1696	0,759168	0,852628	1,147372	Lower than null reference
Ta`if	160	225,4204	0,709785	0,801964	1,198036	Lower than null reference
Qunfudah	38	54,343	0,699262	0,590689	1,409311	Within null reference
Tabouk	116	166,649	0,696074	0,76848	1,23152	Lower than null reference
Jeddah	489	805,6269	0,606981	0,900676	1,099324	Lower than null reference
Hafr Al-Baten	36	84,39708	0,426555	0,672383	1,327617	Lower than null reference
Eastern	254	622,8733	0,407788	0,885046	1,114954	Lower than null reference
Bishah	25	65,90997	0,379305	0,628715	1,371285	Lower than null reference
Ha`il	46	133,1055	0,345591	0,740205	1,259795	Lower than null reference
Al-Ahsa	51	217,4263	0,234562	0,798202	1,201798	Lower than null reference
Al-Bahah	16	77,49369	0,206468	0,657955	1,342045	Lower than null reference

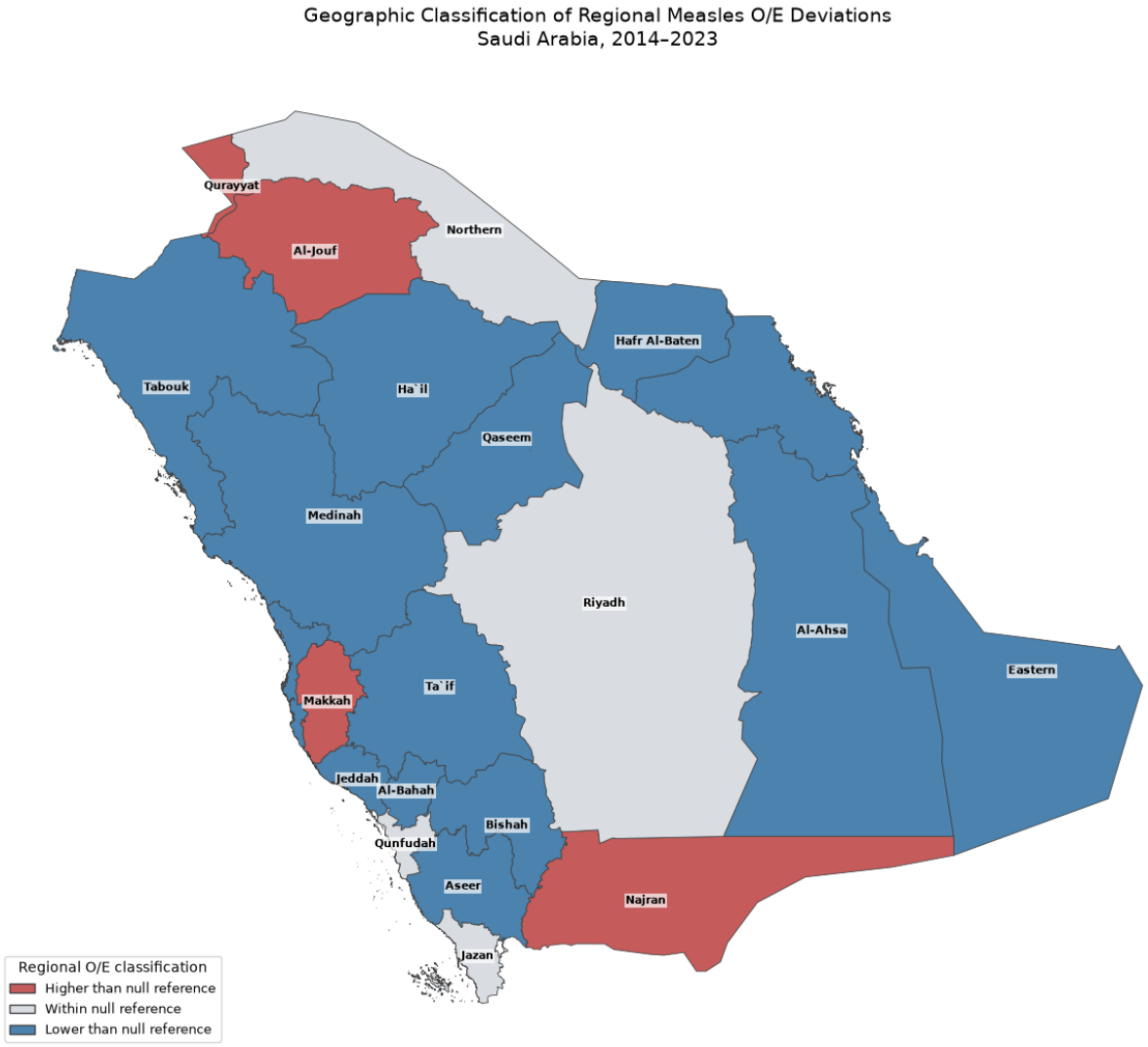

Geographic Classification of Regional Measles O/E Deviations
Saudi Arabia, 2014–2023
Qurayyat
Northern
Al-Jouf
Hafr Al-Baten
Tabouk
Ha`il
Qaseem
Medinah
Riyadh
Al-Ahsa
Eastern
Ta`if
Makkah
Jeddah
Al-Bahah
Bishah
Qunfudah
Aseer
Najran
Jazan
Regional O/E classification
Higher than null reference
Within null reference
Lower than null reference
Simultaneous 95% multinomial null reference; provisional geographic boundaries

Figure 21